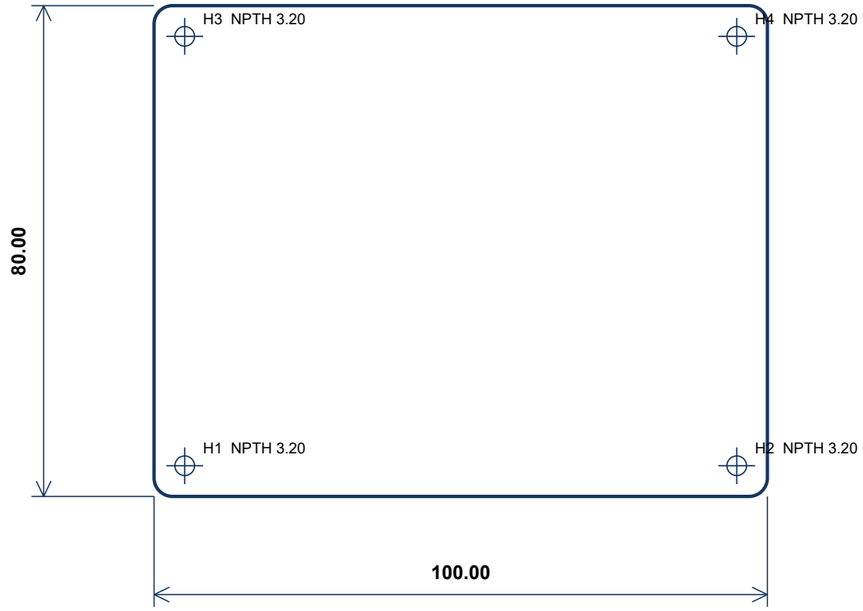
EVT_SYS_CARRIER REV.A | BOARD OUTLINE AND PILOT FABRICATION BASIS

Dimensions in millimetres. Mechanical datum and output status are bound to the final native Altium PcbDoc.

PILOT FABRICATION NOTES

1. Finished board outline: 100.00 x 80.00 mm with R3.00 corners.
2. Project mechanical target: 1.60 mm. Working Rezonit baseline: published 1.50 mm TG150 four-layer construction; order data prevails.
3. Four copper layers, 35 um base copper on every layer, FR-4 TG150 IPC-4101/99 working material.
4. L1 components/signals; L2 solid GND; L3 power and slow signals; L4 secondary signals.
5. Pilot finish request: ENIG, 0.075-0.125 um Au over 3-6 um Ni; green solder mask; white legend.
6. Controlled impedance targets: GNSS_RF 50 ohm and ETH_MDI 100 ohm differential, +/-10 percent target.
7. Rezonit published starting geometry and routed EVT geometry differ. Do not claim controlled impedance without CAM solver review and accepted PCB ECO.
8. Gerber X2 includes board profile and PTH/NPTH drill plots. Legacy RS-274X GKO is retained as a CAM fallback.
9. Use the NC drill report for plated and non-plated tool definitions. Finished hole sizes are specified on sheet 2.
10. Electrical test is required. Submit single-board data; fabricator panelisation is permitted after CAM review.
11. IPC-2581 Revision B and Gerber X2 are both supplied. Resolve any CAM disagreement against the native PcbDoc and this drawing.
12. This pilot drawing supports quotation and CAM preflight. It does not grant production authority.

OUTLINE: R3.00 CORNERS, ROUTED PROFILE



MOUNTING DATUM: H1 CENTRE AT X=5.00, Y=5.00 FROM LOWER-LEFT BOARD EXTENTS

Item	Current evidence	Pilot R1 state
Gerber X2	4 copper, mask, paste, legend, profile, PTH/NPTH	PASS
IPC-2581	Revision B; 92 components, 20 layers, 172 holes	PASS
Native DRC	0 warnings, 0 violations, 0 unrouted	PASS
Connections	529 pads, 1639 tracks, 132 vias, 67 fills	PASS
STEP	Board solid and hole geometry; component 3D out of scope	PASS SCOPE

EVT_SYS_CARRIER REV.A

PILOT R1 FABRICATION DRAWING
FABRICATOR REVIEW / CAM PREFLIGHT

Scale: 1:1

PRODUCTION AUTHORITY: NOT_AUTHORIZED

Sheet 1/2

EVT_SYS_CARRIER REV.A | STACKUP, IMPEDANCE, DFM AND DRILL SUMMARY

Pilot R1 uses the published Rezonit 4-layer, 35 um TG150 construction as a working baseline; order-specific values remain deferred.

PUBLISHED REZONIT WORKING STACK

Seq	Item	Layer / material	Thickness	Copper	Dk at 1 MHz	Status
1	Copper	L1 Top	0.035 mm	35 um		WORKING
2	Prepreg	FR-4 TG150 2116	0.115 mm		4.2	WORKING
3	Copper	L2 solid GND	0.035 mm	35 um		WORKING
4	Core	FR-4 TG150 IPC-4101/99	1.130 mm		4.6	WORKING
5	Copper	L3 power / slow signal	0.035 mm	35 um		WORKING
6	Prepreg	FR-4 TG150 2116	0.115 mm		4.2	WORKING
7	Copper	L4 Bottom	0.035 mm	35 um		WORKING

Published actual board thickness: 1.50 mm, excluding plating, finish and solder mask. Project mechanical target remains 1.60 mm.

CONTROLLED IMPEDANCE COMPARISON

Class	Structure	Target	Rezonit published 1.50 mm start	Routed EVT start	Disposition
GNSS_RF	L1 microstrip to L2	50 ohm	W 0.180 mm	W 0.360 mm; analytical 49.93 ohm	CAM SOLVER + ECO
ETH_MDI	L1 edge-coupled pair to L2	100 ohm diff	W 0.150 / gap 0.300 mm	W 0.260 / gap 0.200 mm; analytical 100.45 ohm	CAM SOLVER + ECO

The website geometry is a starting value, not an order-bound solver result. The supplied artwork remains the validated EVT baseline. Any impedance geometry correction requires a controlled PCB ECO and repeated DRC/connection audit.

NC DRILL TOOL SUMMARY

Tool	Finished hole	Type	Count
T1	0.20 mm	PTH	8
T2	0.25 mm	PTH	5
T3	0.30 mm	PTH	121
T4	0.40 mm	PTH	2
T5	0.90 mm	PTH	8
T6	0.991 mm	PTH	3
T7	1.02 mm	PTH	10
T8	1.10 mm	PTH	4
T9	1.60 mm	PTH	2
T10	3.00 mm	PTH	3
T11	3.25 mm	PTH	2
T12	3.20 mm	NPTH	4
	TOTAL		172

PUBLISHED DFM BASELINE VS CURRENT DESIGN

Parameter	Current design / output	Rezonit published	Pilot disposition
Trace width	0.15 mm minimum rule	0.20 basic; 0.15 advanced	ADVANCED
Copper spacing	0.20 mm minimum rule	0.20 basic; 0.15 advanced	PASS BASIC
PTH drill	0.20 mm minimum in NC output	0.30 basic; 0.20 high complexity	ADVANCED
Via annular ring	0.15 mm for 0.60/0.30 via	0.15 basic; 0.10 advanced	PASS BASIC
NPTH	3.20 mm mounting holes	0.50 mm published minimum	PASS
Solder mask	Green; 0.025 mm expansion request	0.10 mm bridge; 0.025 expansion	CAM CHECK

SOURCE BINDING

Native PcbDoc SHA-256: ec04f945b1e617438fc465fea1b5fa28c55ba87d0374be4110528dbac0a24816
Native DRC: PASS, 0 warnings, 0 violations, 0 unrouted nets.
Connection integrity: PASS; all pads and routed copper remain inside the 100 x 80 mm outline.
Rezonit stack source: rezonit.ru/.../tipovye-sborki-i-konstruktsii-pechatnykh-plat/
Rezonit finish/mask source: rezonit.ru published production capabilities table.
Order-specific stack, finished copper and field-solver geometry take precedence when available.

PRODUCTION AUTHORITY REMAINS NOT_AUTHORIZED

EVT_SYS_CARRIER REV.A
PILOT R1 FABRICATION DRAWING
FABRICATOR REVIEW / CAM PREFLIGHT

Scale: 1:1
PRODUCTION AUTHORITY: NOT_AUTHORIZED